Siraj J. Lyons

Curriculum Vitae

Email: siraj.lyons@louisville.edu

Last Updated: November 13, 2026

EDUCATION

University of Louisville, Louisville, KY | *PhD* **2022 - 2027 (Expected)**
- Area: Psychological Science
- PI: Brendan E. Depue, PhD
- Dissertation: Brainstem nuclei in bipolar disorder: A morphometric and resting-state entropy investigation of the ascending arousal network

Northern Michigan University, Marquette, MI | *MSc* **2020 - 2022**
- Area: Psychological Science
- PI: Lin Fang, PhD
- Thesis: Electrocortical changes following attention bias modification in a sample of high-bipolar trait individuals

University of Toledo, Toledo, OH | *BA* **2016 - 2020**
- Major: Psychology

RESEARCH EXPERIENCE

Graduate Student **2022 - Present**
- PI: Brendan E. Depue, PhD
- University of Louisville, Louisville, KY
- Tasks: Independent research, Mentoring, Asset management

Graduate Student **2020 - 2022**
- PI: Lin Fang, PhD
- Northern Michigan University, Marquette, MI
- Tasks: Independent research, Mentoring

Research Assistant **2018 - 2020**
- PIs: John D. Jasper, PhD; Wesley Bullock, PhD, Matthew Tull, PhD; Kim Gratz, PhD
- University of Toledo, Toledo, OH
- Tasks: Data collection, Documenting protocols, Independent research

MANUSCRIPTS

** Denotes trainees*

Lyons, S. J., Foley, J., Cook, O., & Depue, B. E. (2026). Bioinformed idiographic symptom networks to identify neurobiological predictors of affective-state transitions in bipolar disorder. Imaging Neuroscience, 4. Article number: IMAG.a.1327. https://doi.org/10.1162/IMAG.a.1327

Foley, F., Lyons, S. J., Cook, O., & Depue, B. E. (2026). Investigating the Functional Neural Network Dynamics of Cognitive-Affective Action Planning during Threat Processing. Frontiers in Psychology.

Lawrence, J. M., Foote, I. F., Breunig, S., Schaffer, L. S., Lyons, S., Abramowitz, S., Levin, M. G., Damrauer, S. M., Mallard, T. T., Grotzinger, A. D. (2026). Shared genetic liability across

systems of psychiatric and physical illness. Nature Communications. https://doi.org/10.1038/s41467-026-69218-1

Lyons, S., Beck, I.*, & Depue, B. E. (2025). Depression is marked by differences in structural covariance between deep brain nuclei and sensorimotor cortex. NeuroImage, 310. https://doi.org/10.1016/j.neuroimage.2025.121127

Lyons, S. & Depue, B. E. (2023). Not all bad decisions are alike: Approach and avoidant bad decisions are associated with distinct network organization. Frontiers in Neuroscience, 17. Article number: 1249008. https://doi.org/10.3389/fnins.2023.1249008.

Submitted

Lawrence, J.M., Breunig, S., Schaffer, L. S., Sheppard, A., Zorina-Lichtenwalter, K., Lyons, S. J., & Grotzinger, A. D. (Under Review, R1). Genetic signal augmentation of childhood-onset and treatment-resistant major depression reveals distinct biological disorders. (*Note* Authorship added during review process.)

Lyons, S. J., & Depue, B. E. (Initial Review). Sensorimotor inference as an adjunctive model of bipolar disorder.

In Preparation (descending order of progress)

Lyons, S. J., Saki, J., Reid, A., Christman, S. D., & Jasper, J. D. (In prep). Unilateral gaze and handedness interact to influence Iowa Gambling Task performance across the hypomanic personality spectrum. Target Journal: Laterality.

Lyons, S. J., Foley, J. F., Gruner, S.*, & Fang, L. (In Prep). Personality clusters predicting motivational biases in two independent tasks: A data-driven approach. Target Journal: Personality and Individual Differences.

Lyons, S. J., Beck, I., Kolks, Samantha*, Komula, Bruce*, Foley, John, & Depue, B. E., The Depression Imaging Consortium. (In Prep). Spectral properties of resting-state sensory and frontal cortical activity distinguish MDD from unaffected individuals. Target Journal: Human Brain Mapping.

GRANTS & AWARDS

Dissertation Completion Award **2027**
University of Louisville
Intramural award for completing the dissertation in program's final semester.

Dean's Citation Award **2024**
University of Louisville
Award presented to a graduating Master's or Doctoral student whose work excelled in research, teaching, or service beyond a high-grade point average. Award given for Master's degree.

Excellence in Research (Junior Level) **2024**
University of Louisville
Department award recognizing research trainees who have excelled in research who have yet to enter doctoral candidacy

Travel Award **2023**
Granting Institution: University of Louisville $350 award for presenting research at a national conference

University Fellowship **2022 - 2024**
University of Louisville
Two-year fellowship awarded to high-ranking incoming doctoral students

Dean's Research Award **2022**
Granting Institution: Northern Michigan University $500 intramural award funding summer research

Excellence in Education **2021**
Granting Institution: Northern Michigan University $1,500 intramural grant funding summer research.

SUPERVISION

University of Louisville
- Samantha Kolks
- Bruce Komula
- Jake Cox
- Seth Gruner
- Sasha Domenech
- Isak Beck
- Jackson Bower
- Zachary Irvin

Northern Michigan University
- John Foley
- Haley Gaboury

University of Toledo
- Alex Heid

PRESENTATIONS

Lyons, S. J. (August, 2026). Resting-state fMRI variability (modestly) predicts Major Depression Disorder. University of Louisville. Program Colloquium.

Lyons, S. (September, 2025). Ascending arousal network morphometry and Big 5 personality traits: A data-driven analysis. University of Louisville. Program Colloquium.

Lyons, S. (April, 2025). Deep-brain nuclei structural covariance with sensorimotor cortex is associated with major depressive disorder. Invited Talk. Northern Michigan University. Declined (funding shortfall).

Lyons, S. (February, 2025). Deep-brain nuclei and Major Depressive Disorder: What's the connection? University of Louisville. Program Colloquium.

Lyons, S. (February, 2024). A brief survey of reward in the brain: Future Avenues. University of Louisville. Program Colloquium.

CONFERENCE PRESENTATIONS

Lyons, S. J., Beck, I., Foley, J., & Depue, B. E. (November, 2026). Resting-state fMRI entropy is heterogenous across major depressive disorder and is linked to clinical trajectories. Poster for Society for Neuroscience. Washington D.C.

Foley, J., Lyons, S. J., & Depue, B. E. (November, 2026). Fixel-based white matter differences across the lifespan and their relationship to trust. Poster for Society for Neuroscience. Washington D.C.

Lyons, S., Beck, I., & Depue, B. (March – April, 2025). Structural co-dependence between deep brain nuclei and sensorimotor cortex differs between people with depression and unaffected individuals. Poster for Cognitive Neuroscience Society.

Lyons, S., Irvin, Z*., & Depue, B. (November, 2023). Frontoparietal network stability predicts attentional control, not cognitive flexibility. Poster for Society for Neuroscience.

Cook, O., Lyons, S., Depue, B. (March, 2023). Cognitive and Motor Neural Network Interaction during Threat Response. Poster for Cognitive Neuroscience Society.

Lyons, S. & Depue, B. (March, 2023). Behavioral approach is linked to increased neural communication between frontostriatal and insular regions during risky decision-making. Poster for Cognitive Neuroscience Society.

Lyons, S., Carlson, J. M., & Fang, L. (November, 2022). Frontal and striatal volumes at baseline predict changes in anxiety symptomology following attention bias modification. Poster for Society for Neuroscience.

Gentry, L., Lyons, S., Kassel, D., Davis, B., Kwak, B., Carlson, J. M., & Fang, L. (November, 2022). Reduced intrinsic connectivity between the dorsal anterior cingulate and medial prefrontal cortices predicts anxiolytic benefits of attention bias modification training in high trait anxious individuals. Poster for Society for Neuroscience.

Hauler, A., Foley, J., Kassel, D., Lyons, S., Caughtry-Carpenter, C., Fang, L., Carlson, J. (September-October, 2022). Reliability of Eye Tracking Based Measures of Attentional Bias in the Scrambled Sentence Task. Poster for Society for Psychophysiological Research.

Kassel, D., Hauler, A., Lyons, S., Gaboury, H., Girard, C., Carlson, J. M., & Fang, L. (April, 2022). Shared and Unique Patterns of Resting-State Functional Connectivity Linked to Attentional and Interpretation Biases. Poster for Cognitive Neuroscience Society.

Lyons, S., Gaboury, H.*, Foley, J.*, Delgado, X., Hauler, A., Carlson, J. M., & Fang, L. (April 2022). Changes in attention and reward-sensitivity ERPs following attention training in a sample of bipolar disorder susceptible individuals. Poseter for Cognitive Neuroscience Society.

Lyons, S., Gilbertson, H., Girard, C., Carlson, J. M., & Fang, L. (November, 2021). Prefrontal cortex based intrinsic functional connectivity is related to electrocortical measures of conflict

monitoring. Poster for Society for Neuroscience.

SERVICE

Preliminary Exam Q&A **2026**
Served as a panelist for the program Question and Answer session for pre-candidacy PhD students.

Pursuing Liberty, Unity, and Support (+) Committee **2025 - *Present***
Graduate Student Member of Reports Subcommittee.

Graduate Network in Arts & Sciences Representative **2025 - 2026**
Chair of the Academic and Grants committee.

Peer Mentor **2024 - *Present***
Point of contact for a 1st year graduate student within the Psychological Science program.

TEACHING EXPERIENCE

Graduate Teaching Assistant | University of Louisville **2024 - Present**

- PSYC 601: Research and Professional Development Seminar; N = 16
  - Led scheduling for department tasks
  - Fall 2026 - Spring 2027
- PSYC 305: Brain and Behavior; N = 100
  - Assisted development of instructional material, quizzes, and exams.
- PSYC 381: Cognitive Neurosience; N = 60
  - Assisted grading and short-essays.
  - Gave lecture on *Emotion Neuroscience* under supervision of instructor of record (Spring 2025)
  - Expected to give lecture on Consciousness (Fall 2026)
  - Spring 2025, Fall 2026
- PSYC 301: Pyschological Statistics; N = 100
  - Developed lab materials for practicing core principals.
  - Led lab instruction.
  - Fall 2025, Spring 2026

Graduate Teaching Assistant | Northern Michigan University **2020 - 2022**

- PSY 100: Introduction to Psychology
  - Led lab instruction for two sections (N = 28)
  - Fall 2020 - Spring 2022
- PSY 250: Brain and Behavior (N = 20)
  - Led lab instruction
  - Trained graduate student on teaching protocols
  - Fall 2020 - Spring 2022

Undergraduate Teaching Assistant | University of Toledo **2020**

- Research Methods (N = 13)
  - Assisted primary instructor and graduate teaching assistant.
  - Graded exams and reviewed research reports.
  - Spring 2020

METHODOLOGY

- EEG acquisition and analysis
  - HydroCel Geodesic
  - MNE
- MRI analysis
  - FSL
  - CONN
  - SPM
  - FreeSurfer
- Behavioral task design
  - PsychoPy
  - EPrime3
  - jsPsych

CODING

- R
- MATLAB
- Bash scripting
- LaTeX (Manuscript preparation)
- Python